

Simple Low-Noise AUDIO-DISTRIBUTION AMPLIFIER

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This article describes a low-noise, low-cost audio-distribution amplifier based on hybrid or mixed technology. Input to the amplifier is from low-noise, high-gain npn transistor. Output buffers are implemented with voltage followers using an operational amplifier (op-amp). The combination of low-noise transistor and push-pull output stage of the op-amp guarantees good characteristics of the hybrid amplifier.

Often we are required to send signal from one audio source to many loads/speakers. For that we use audio-distribution amplifiers. This circuit employs low-noise bipolar transistor BC550 and op-amp LM324, which are widely available

at low cost.

The circuit design involves the following:

Buffers. The individual output buffers/followers of the op-amp work as push-pull output drivers separating the outputs from each other and the preamplifier from the loads.

Preamplifier. Common low-noise preamplifier using BC550 guarantees low noise in the entire circuit. Common negative AC and DC feedback for the preamplifier and the first output buffer guarantee low total harmonic distortion.

Gain. Gain in the preamplifier can be set from one to more than 50 depending on the gain of the transistor and the resistors around it.

Common feedback of the pream-

plifier and the output buffer reduces the distortions of LM324.

The methods mentioned above are effective especially when all outputs of the amplifiers are of the same type and are loaded with the same or similar loads.

Circuit and working

Fig. 1 shows the circuit of the low-noise audio-distribution amplifier with four output buffers. Op-amps should be of the same type and preferably in the same package. Resistor R1 and capacitor C2 are used to reduce RFI and EMI. Input amplifier stage is built around transistor T1 (BC550).

R14, C3 and C4 form filter circuit of the power supply for T1. Main

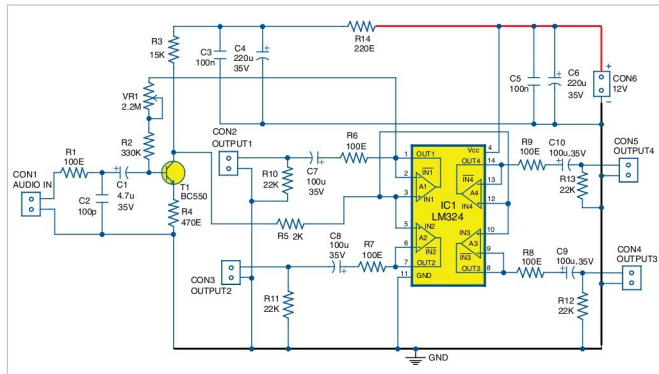


Fig. 1: Circuit diagram of the low-noise audio-distribution amplifier